Engineers decide. AI assembles and reasons over evidence. Rules enforce. The model never holds write authority.

INTAKE

- **Fault Injector** — UI throws a shaped fault
- **Fleet simulator** — 1 Hz telemetry, seeded
- **Operator report** — POST /api/incidents
- **QMS NCR** — Nonconformance event
- **Supplier notice** — Material lot signal

telemetry

DETERMINISTIC — code, unit tested

- **SPC windows** — Western Electric rules 1-4
- **Ticketing** — Alert opens a TKT
- **Intake** — Validate, dedupe, severity
- **Lifecycle** — Only legal transitions
- channels
- **Impact features** — Scope and timing, passed to the agent as features
- to orchestrator

READ-ONLY EVIDENCE

- **MES** — Runs, lots, recipes
- **QMS** — NCRs and CAPAs
- **PLM** — Configs, BOM, ECOs
- **ERP** — Lots and suppliers
- **CMMS** — PM and calibrations
- **Ticketing** — TKT records
- **Corpus** — SOP, FMEA, bulletins, CoA, decoy, injection
- **Chunk and embed** — Heading-aware chunks, versioned index
- **Hybrid retrieval** — BM25 plus vector, family and revision filters

read-only

JUDGMENT PATH

- **Orchestrator** — MAF behind IInvestigationWorkflow
- **Investigation Agent** — Read-only tools, at most 10 calls
- **Evidence gate** — Gaps escalate. No root cause asserted.
- **Resolution Agent** — No tools. Category, citations, confounders
- **Grounding validator** — Cited IDs must be in the pack
- **React console** — The engineer decides
- **Approval token** — HMAC, single-use, bound
- **Action service** — CAPA write. Not an agent tool.
- **Mock QMS** — Creates the CAPA, or a forced 500
- **Model gateway** — Route, retry, breaker, and a $2 per day hard stop
- **Azure OpenAI** — gpt-4o-mini and gpt-4o
- **OpenAI** — Same gateway route
- **Replay** — Over cap, or local default
- **Mock reasoner** — No recording for this fault
- **Audit log** — Append-only. SPC, workflow, gateway.
- **Eval runner** — 24 cases. Hard gates on unsafe answers.
- **Red-team** — Injection doc and operator text
- **Registry** — Prompt, model, and index versions

LEGEND

- Deterministic rules. Code, unit tested.
- Probabilistic agents. No write tools.
- Human authority. The only write path.
- Read-only evidence, audit, eval.
- Model gateway and the spend cap.
- Arrow: a handoff or a read.

MERIDIAN PRECISION SYSTEMS
Quality and Engineering Issue Resolution

Logical architecture
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